TECHNICAL EVALUATION — PLASTIC TUBE WELDING TOOLS; 220–240 V

1. Evaluation output

Field	Value
Report	1.Plastic tube welding tools.pdf (43 pages)
Standards	IEC 60335-2-45-01:2002 (Part 2 2-45-01)
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2. Report identity

Field	Value
Report number	TR22061969-S-001
Laboratory	Guangdong Keyway Testing Technology Co., Ltd.
Report date	2026-6-19
Product declared	Plastic tube welding tools; 220–240 V, 50/60 Hz, 800/1500 W
Declared model	PTWT215002
Standards claimed	US IEC 60335-2-45:2002; US IEC 60335-1:2010
Standards evaluated	US IEC 60335-2-45:2002, supplied particular part; US IEC 60335-1:2010 not supplied

3. Summary

Field	Value
Final verdict	REJECT — INSUFFICIENT TECHNICAL EVIDENCE
Reason	The report uses machine-equipment content rather than a traceable appliance-standard assessment and lacks essential heating, abnormal-operation and mechanical-strength evidence. The claimed general part was not supplied, and the product's scope and welding method require clarification.
Recommended action	Obtain the applicable general standard and amendments, establish product classification and scope, and provide a replacement appliance-specific report with the necessary test data before certification proceeds.
Conclusion	Product compliance has not been demonstrated. The recorded 5 MΩ insulation resistance and 1,000 V electric-strength result do not substitute for the missing appliance-standard assessment; no product non-conformity has been established from the supplied evidence.
Findings	13 total — Critical 2, High 9, Medium 2, Low 0

4. Findings

Priority	Clause	Normative requirement	Finding reason	Required action	Pages	Evidence
Critical	1; 3.115	The standard covers portable electric heating tools within its scope, while exclusively industrial appliances are excluded; a thermoplastic conduit-welding tool uses a fitting with an integral heating wire.	The report calls the product a plastic tube welding tool but also describes a machine located in a factory. Its intended use and welding method are not sufficiently documented to establish scope or whether the fitting-specific requirements apply.	Document intended use and welding method, and establish the applicable product classification and standard scope.	PDF pp.2,8,43	Report: verified p.8 Standard: verified in IEC 60335-2-45-01:2002 p.13
Critical	4; 32	The particular part supplements or modifies the corresponding clauses of IEC 60335-1.	The claimed appliance-standard report contains machine-equipment clauses and an IEC 60204 annex instead of a traceable clause-by-clause appliance assessment; identical clause numbers cannot be treated as evidence of the corresponding appliance tests.	Obtain a replacement report using the applicable appliance-standard requirements and product-specific test records.	PDF pp.1,5–42	Report: verified p.41 Standard: verified in IEC 60335-2-45-01:2002 p.8
High	6.1	Appliances other than dehorning tools shall be class I, II or III.	The report's clause 6.1 concerns machine shock protection and does not identify the tested appliance's protection class.	Record and substantiate the appliance's protection class.	PDF pp.8–9	Report: verified p.8 Standard: verified in IEC 60335-2-45-01:2002 p.14
High	11.2; 11.7	Heating tests are performed away from test-corner walls and, for appliances other than firelighters, continue until steady conditions are established.	The report's clause 11 addresses electronic equipment; no appliance heating test, arrangement, steady-state record or temperature results are provided.	Perform and document the applicable heating assessment and test conditions.	PDF p.24	Report: no quote given Standard: verified in IEC 60335-2-45-01:2002 p.16
High	19.2; 19.3	The abnormal-operation tests use the specified clause 11 conditions at 0.94 and 1.06 times rated voltage, subject to the stated exceptions.	Report clauses 19.2 and 19.3 address protective bonding and insulation resistance, not abnormal operation at the specified voltages.	Conduct and document the applicable abnormal-operation tests at both specified voltages.	PDF pp.39–42	Report: no quote given Standard: verified in IEC 60335-2-45-01:2002 p.17
High	21.1; 21.101	Hand-held appliances are subjected to the specified drop tests from 1 m onto a 50 mm hardwood base.	The photographed tool is hand-held, but the report provides no drop-test procedure or post-test assessment of damage and access to live parts.	Perform and document the hand-held drop tests and post-test examination.	PDF p.43	Report: no quote given Standard: verified in IEC 60335-2-45-01:2002 p.18
High	20.1; 22.101	Applicable hand-held appliances require a stand unless the specified clause 11 exception applies, and stability testing is performed on their stands.	The report does not establish the intended working location, stand provision, applicability of the exception or stability-test result.	Document the intended use and stand arrangement, and assess construction and stability as applicable.	PDF pp.2,43	Report: no quote given Standard: verified in IEC 60335-2-45-01:2002 p.19
High	25.15	The modified cord-anchorage pull force and torque provisions apply to hand-held appliances.	The report does not record appliance mass or an appliance cord-anchorage test, so the applicable test parameters and result cannot be verified.	Record the appliance mass and perform the applicable cord-anchorage assessment.	PDF pp.5–42	Report: no quote given Standard: verified in IEC 60335-2-45-01:2002 p.21
High	30.2	For appliances other than contact firelighters, clause 30.2.2 of Part 1 applies.	No appliance heat-and-fire resistance assessment is recorded; the detailed Part 1 requirements are unavailable because that standard was not supplied.	Obtain the applicable Part 1 and document the required resistance-to-heat-and-fire assessment.	PDF pp.5–42	Report: no quote given Standard: verified in IEC 60335-2-45-01:2002 p.22

Priority	Clause	Normative requirement	Finding reason	Required action	Pages	Evidence
High	8; 10; 13; 14; 15; 16	The corresponding Part 1 clauses apply, subject to the particular-part modifications.	Appliance-specific evidence for access to live parts, power input, leakage current, electric strength, transient overvoltages and moisture resistance is not established by the machine-equipment entries.	Obtain US IEC 60335-1:2010 and its applicable amendments, and provide a clause-by-clause appliance assessment with test data.	PDF pp.5–42	Report: no quote given Standard: verified in IEC 60335-2-45-01:2002 p.12
High	23; 24; 25; 26; 27; 28; 29; 31; 32	The corresponding Part 1 clauses apply, subject to the particular-part modifications.	The report does not provide a traceable appliance assessment of wiring, components, supply connection, terminals, earthing, connections, insulation distances, rusting and radiation or similar hazards.	Provide the applicable Part 1 and a product-specific assessment of these clauses, including measurements where required.	PDF pp.5–42	Report: no quote given Standard: verified in IEC 60335-2-45-01:2002 p.12
Medium	1	Scope and applicability must be established from the supplied product standard.	The report states both that no neutral conductor is used and that a neutral is used, leaving the recorded electrical construction inconsistent.	Reconcile the neutral-conductor statements against the tested sample and correct the report.	PDF pp.6,10,31	Report: verified p.6 Standard: verified in IEC 60335-2-45-01:2002 p.11
Medium	1	The assessment must relate to the product within the stated standard scope.	Page 3 is identified only as "001" rather than the complete report number, weakening page attribution.	Correct page 3 attribution and verify the integrity of the complete report.	PDF p.3	Report: verified p.3 Standard: verified in IEC 60335-2-45-01:2002 p.11